



```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: df=pd.read_csv("iris.csv")
```

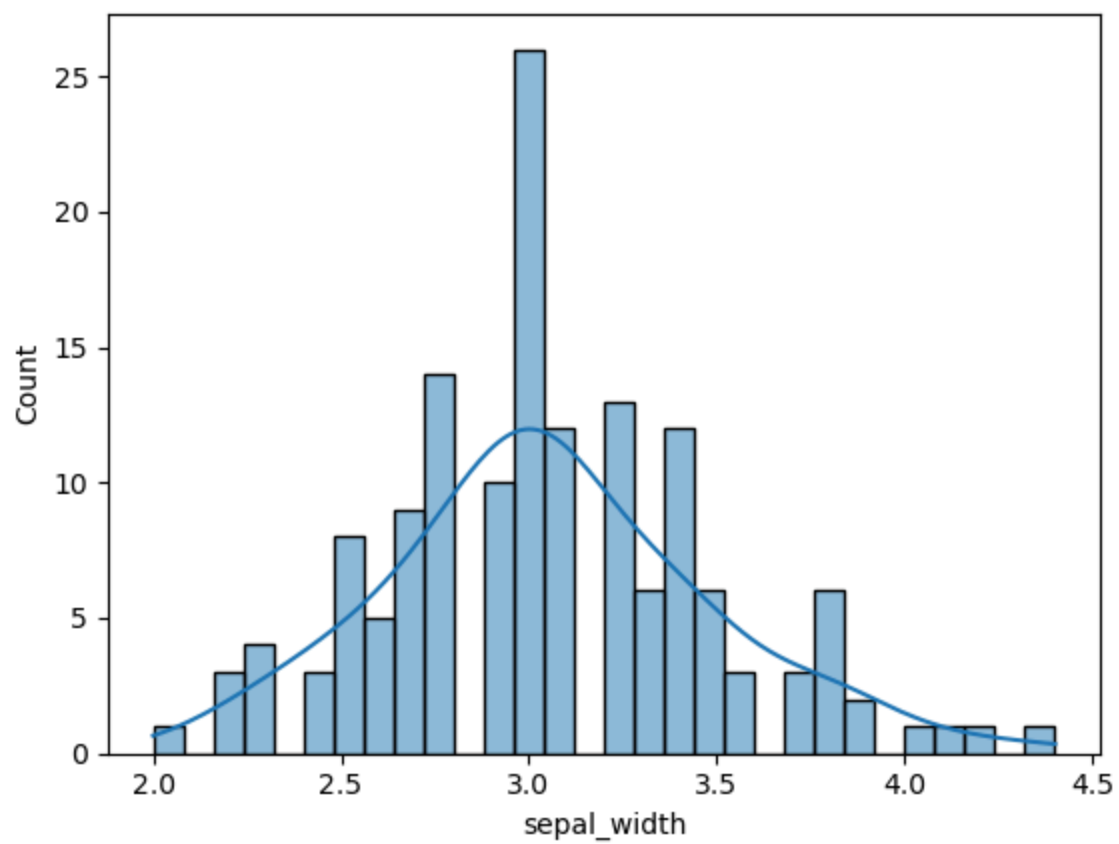
```
In [5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

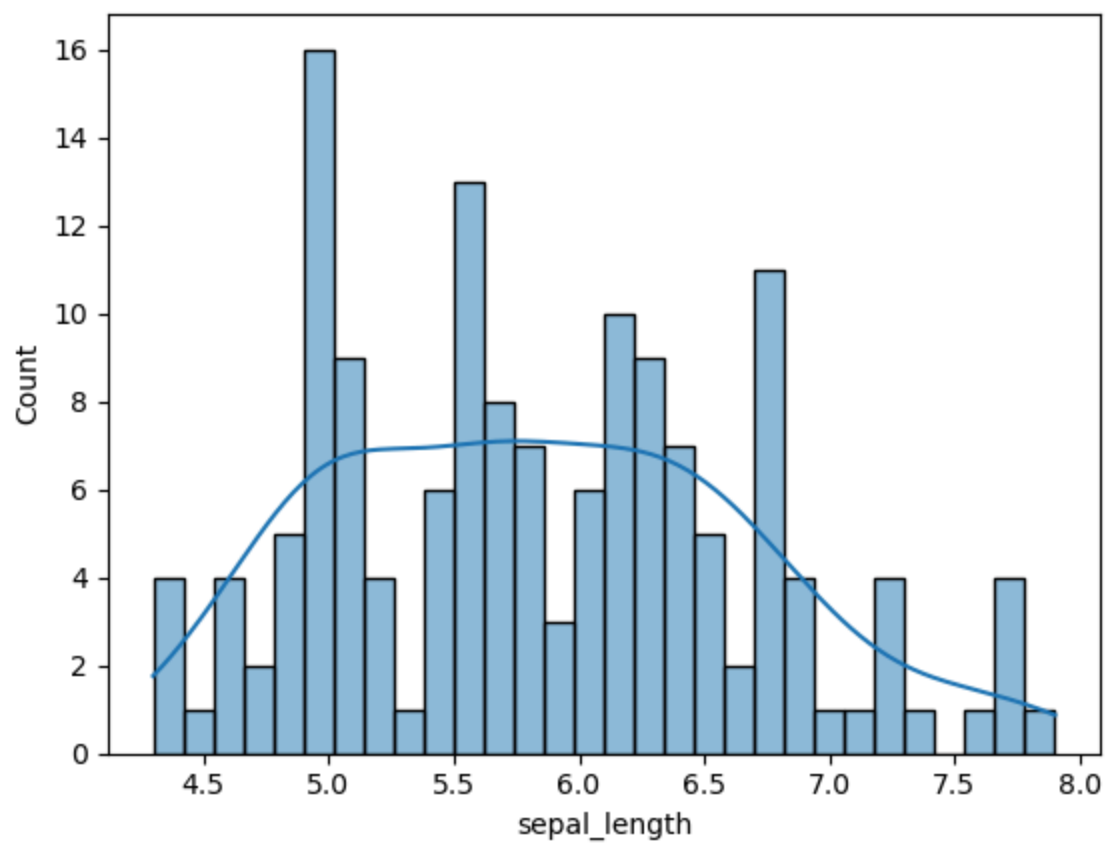
```
In [7]: df.columns
```

```
Out[7]: Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
              'species'],
              dtype='object')
```

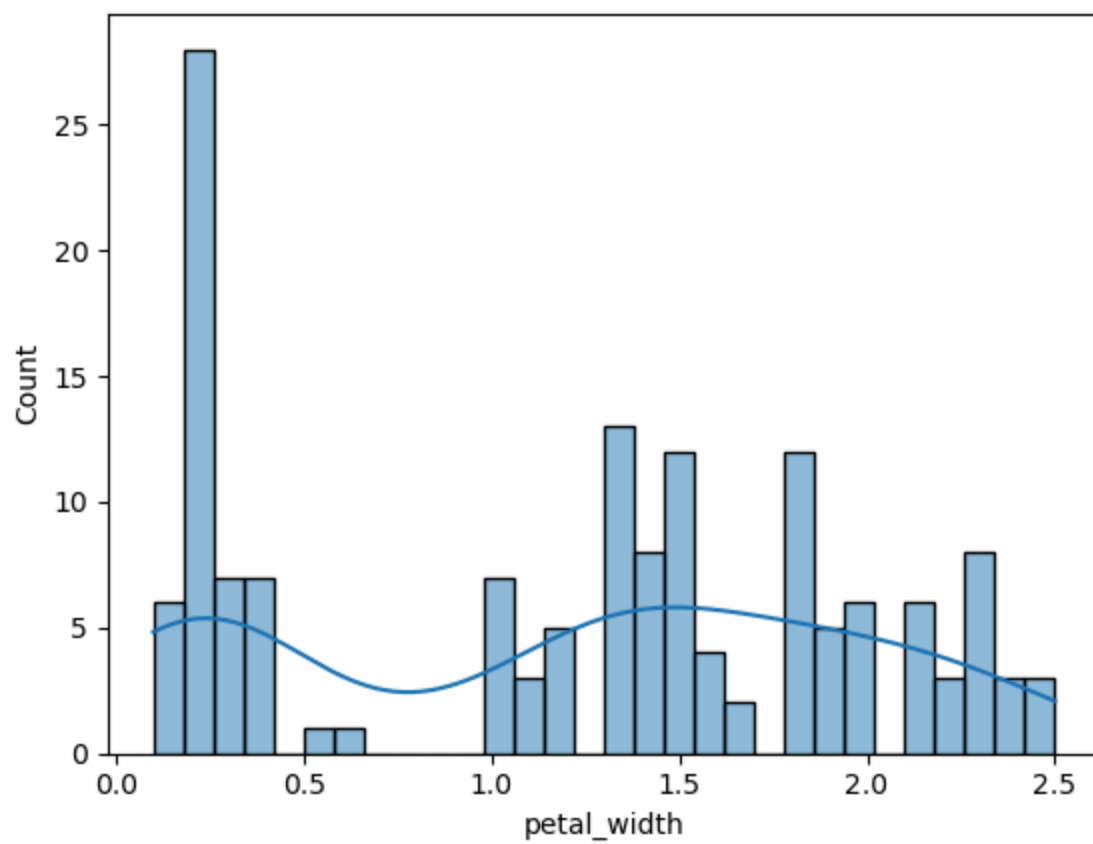
```
In [20]: sns.histplot(df['sepal_width'], bins=30, kde=True)
plt.show()
```



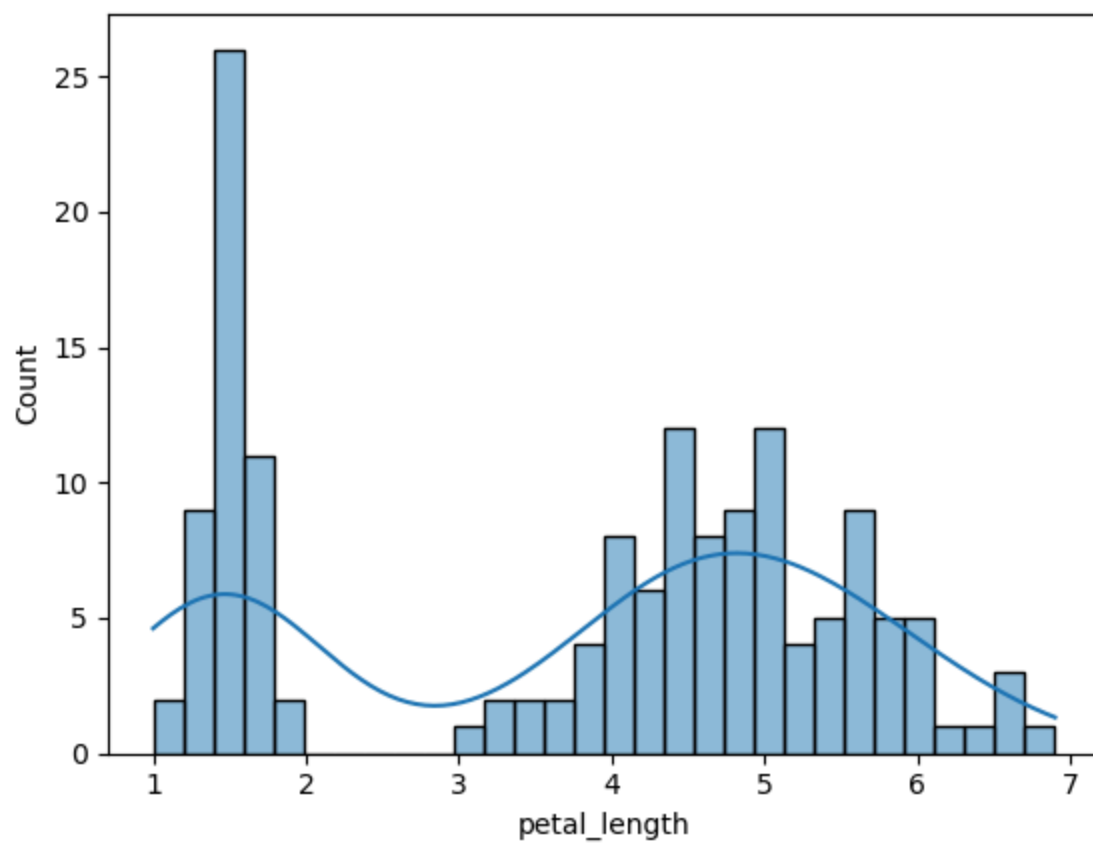
```
In [21]: sns.histplot(df['sepal_length'], bins=30, kde=True)
plt.show()
```



```
In [23]: sns.histplot(df['petal_width'], bins=30, kde=True)
plt.show()
```

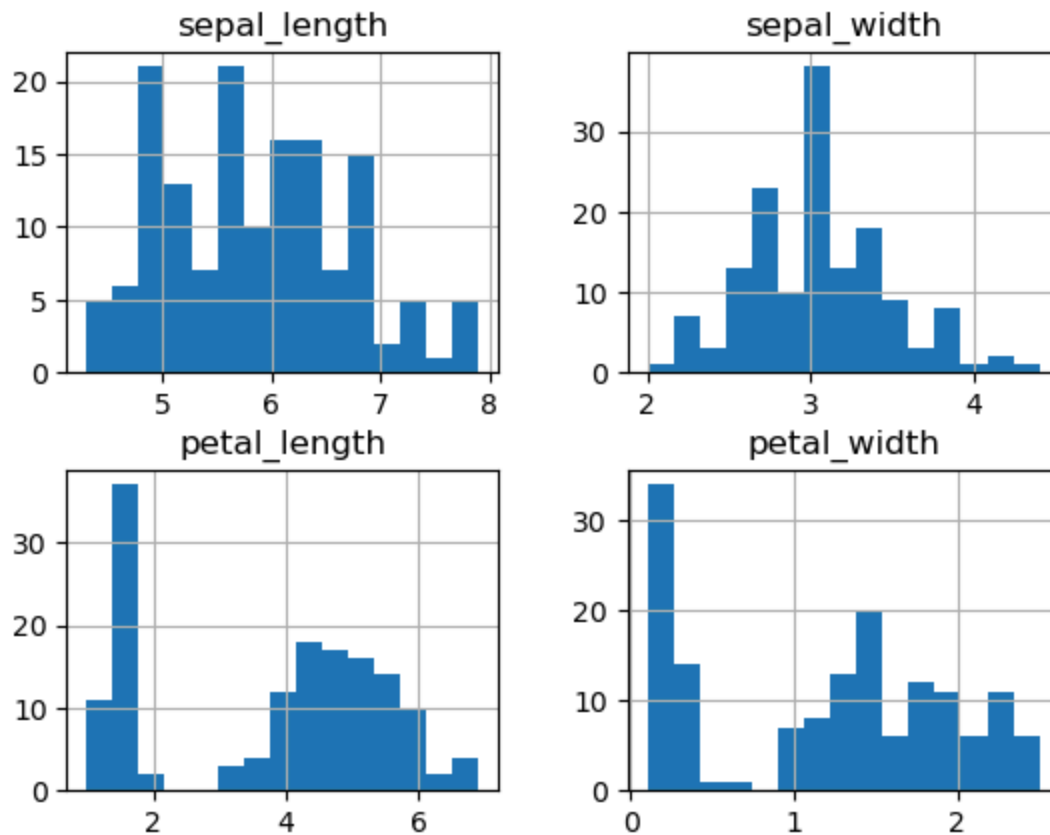


```
In [24]: sns.histplot(df['petal_length'], bins=30, kde=True)
plt.show()
```



```
In [18]: df.hist(bins=15)
plt.suptitle('Histograms of Iris Features')
plt.show()
```

Histograms of Iris Features



Inference: Histograms (Feature Distributions)

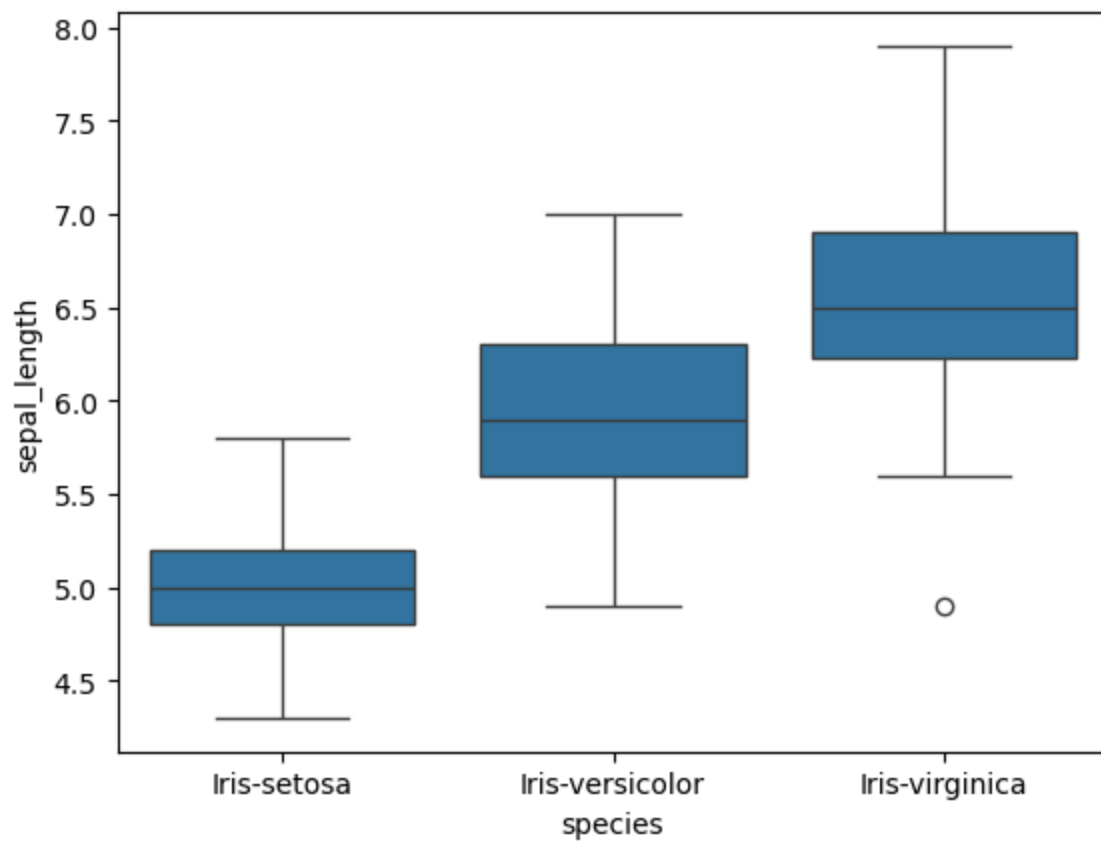
Sepal Length: Shows a relatively normal distribution with a slight right skew and peaks between 5.5 cm and 6.5 cm.

Sepal Width: Closely follows a Gaussian (normal) distribution, peaking symmetrically around 3.0 cm.

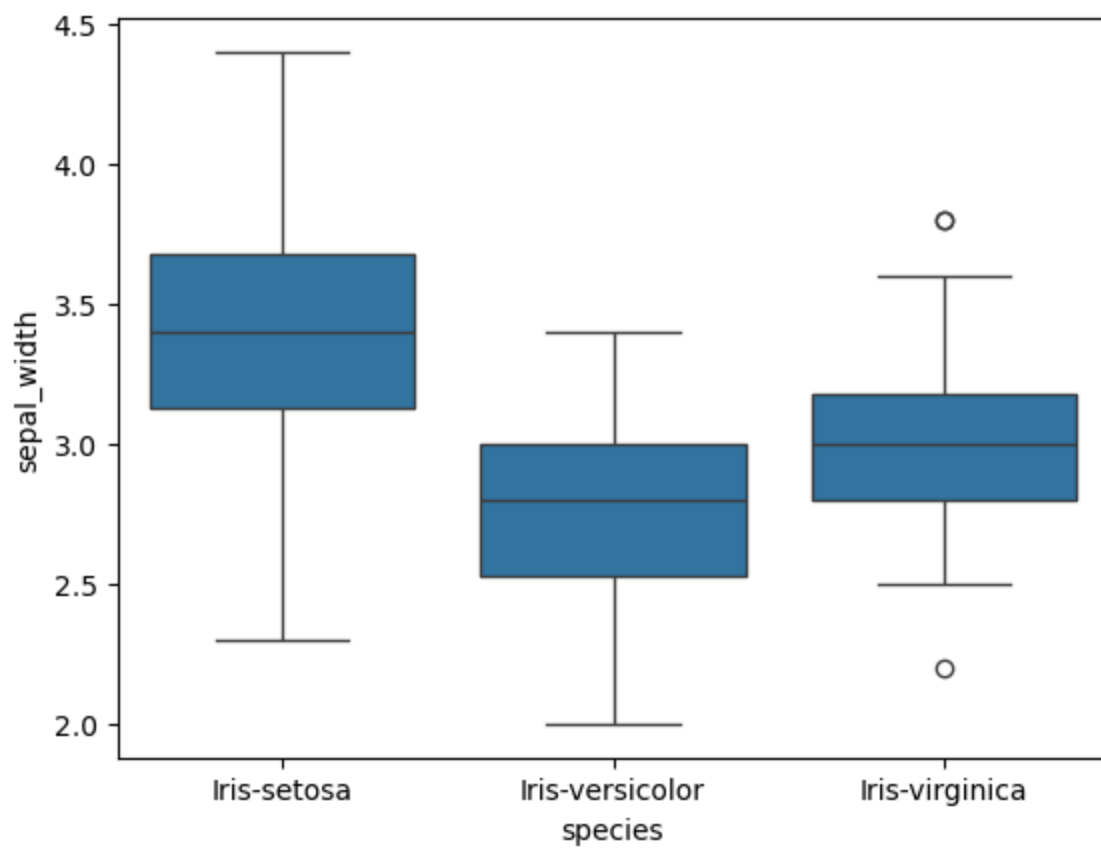
Petal Length: Exhibits a distinct bimodal distribution with a large gap, indicating two separate groups of flowers.

Petal Width: Also displays a bimodal distribution, mirroring the clear separation between species seen in petal length.

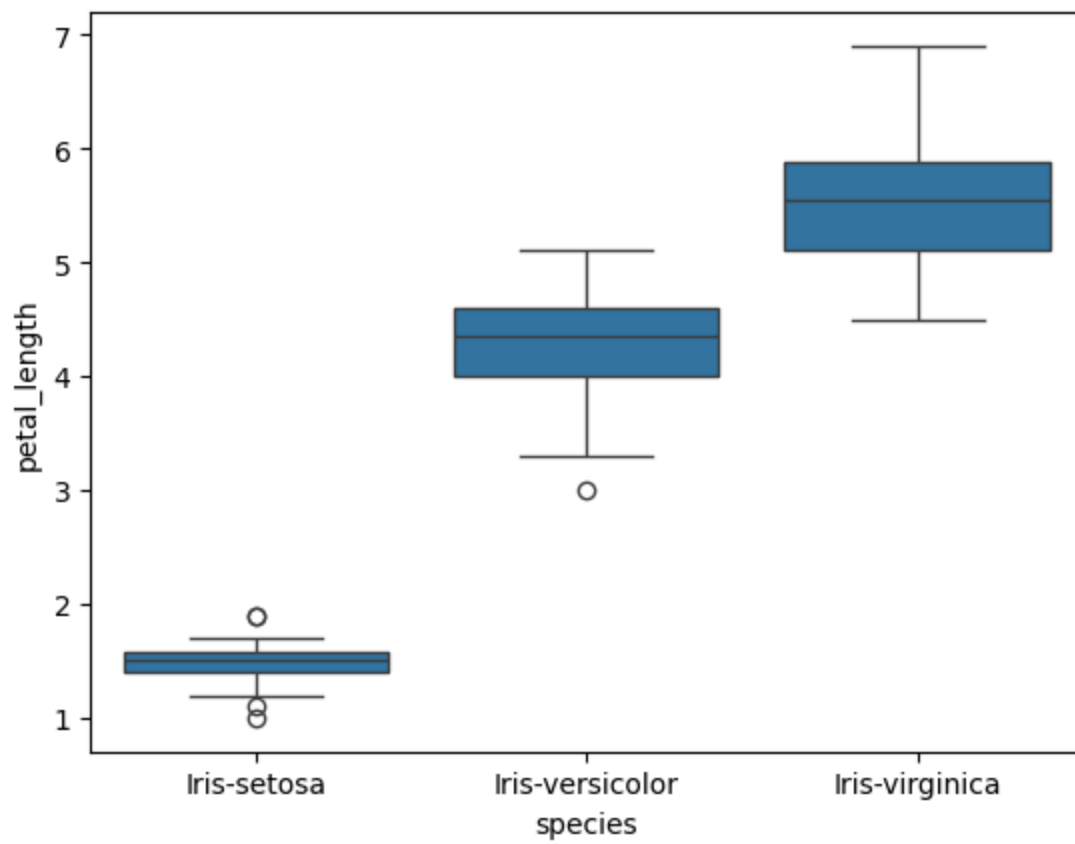
```
In [25]: sns.boxplot(x="species", y="sepal_length", data=df)
plt.show()
```



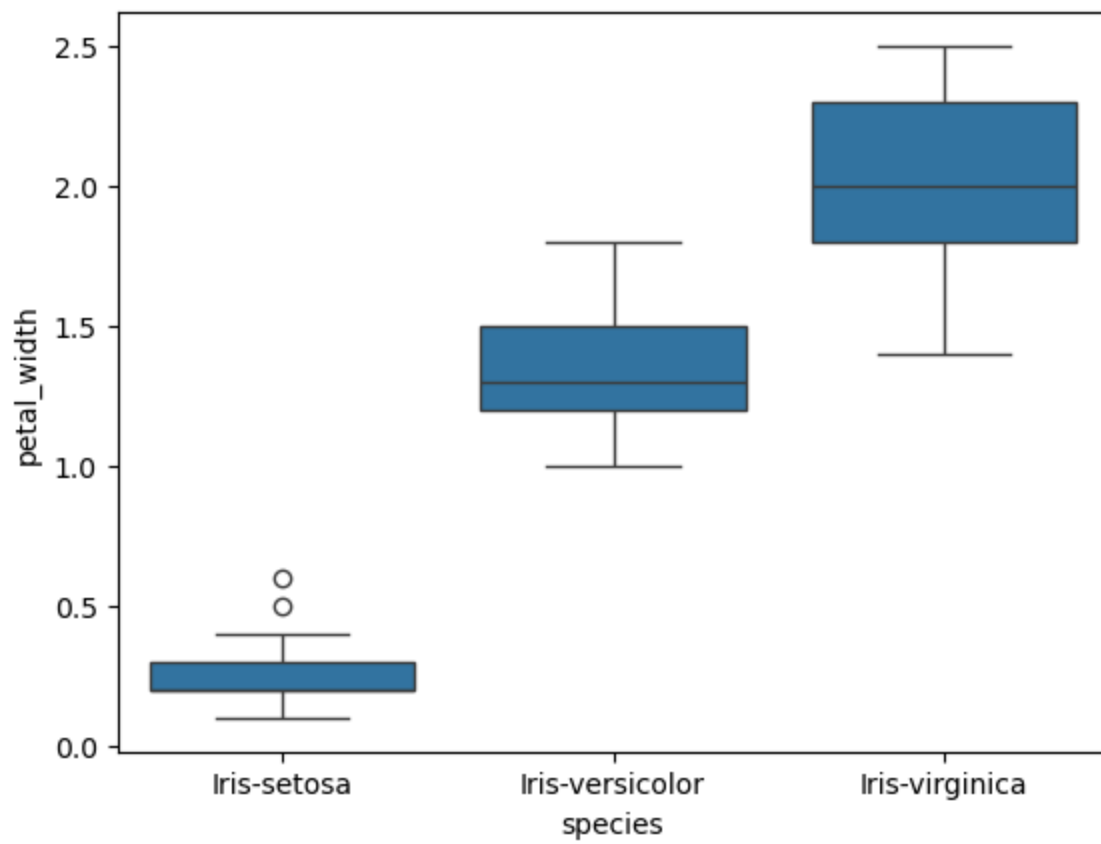
```
In [26]: sns.boxplot(x="species", y="sepal_width", data=df)
plt.show()
```



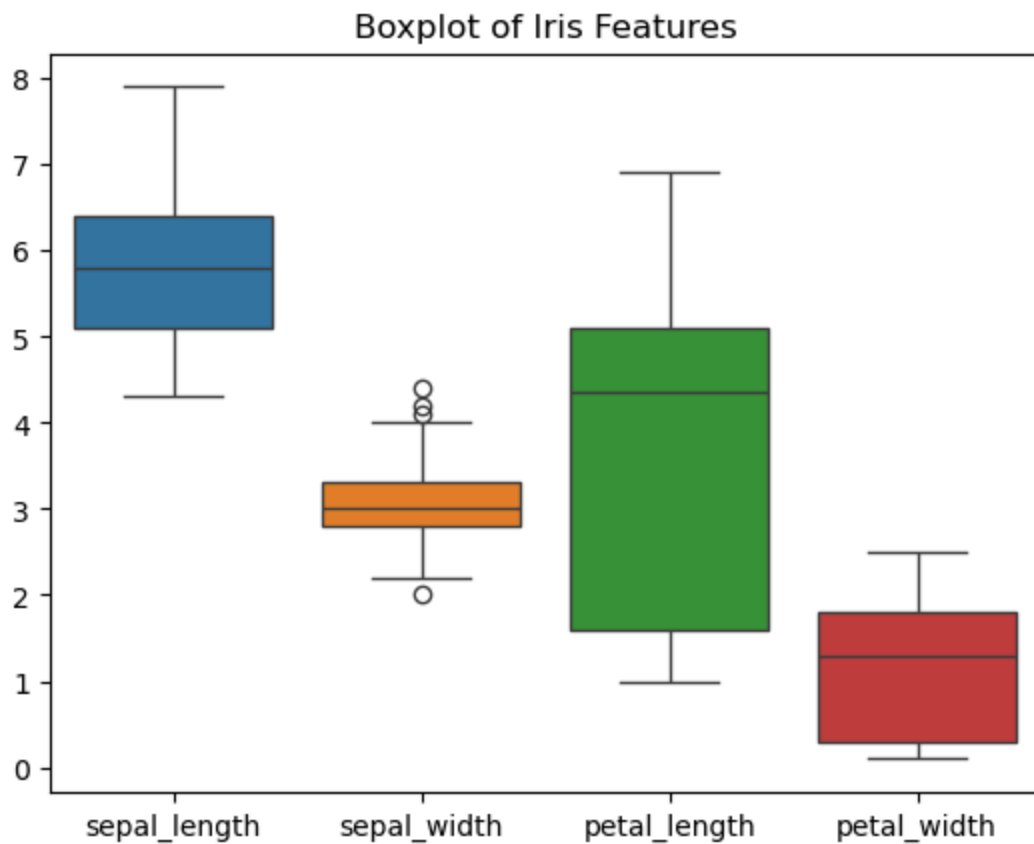
```
In [29]: sns.boxplot(x="species", y="petal_length", data=df)
plt.show()
```

```
In [30]: sns.boxplot(x="species", y="petal_width", data=df)
plt.show()
```



```
In [14]: sns.boxplot(data=df)
plt.title('Boxplot of Iris Features')
plt.show()
```



Inference: Boxplots (Spread and Outliers)

Sepal Length: Shows a moderate interquartile range with a median centered toward the lower half and no outliers.

Sepal Width: Features a narrow spread and is the only variable containing significant outliers beyond both the upper and lower whiskers.

Petal Length: Displays the largest variation in data spread (IQR) across the dataset but remains free of outliers.

Petal Width: Shows a wide distribution relative to its small values with a median shifted toward the bottom and no outliers.